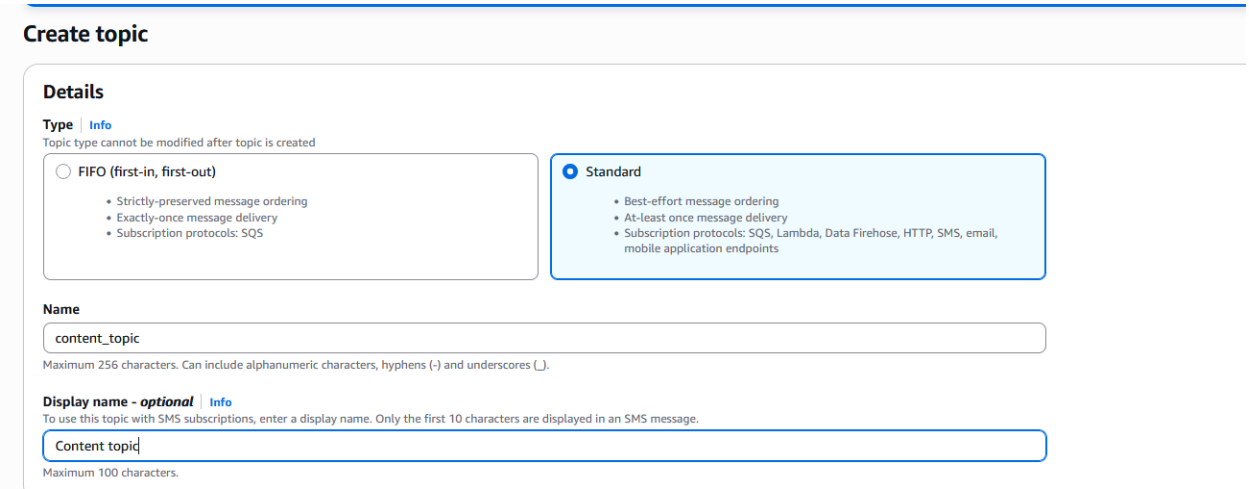
AWS SNS and SQS Integration Project

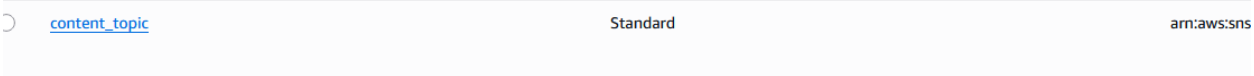
Project Overview

This section details the step-by-step process for integrating Amazon Simple Notification Service (SNS) with Amazon Simple Queue Service (SQS). The goal was to establish a publish-subscribe messaging workflow, allowing SNS to fan out messages to SQS queues for asynchronous processing.

Step 1: SNS Topic Creation

The integration process began with the creation of an SNS topic. Using the SNS Management Console, a Standard topic type was selected and named content_topic. The display name assigned was "Content topic." The topic was created successfully, establishing the foundation for sending notifications.



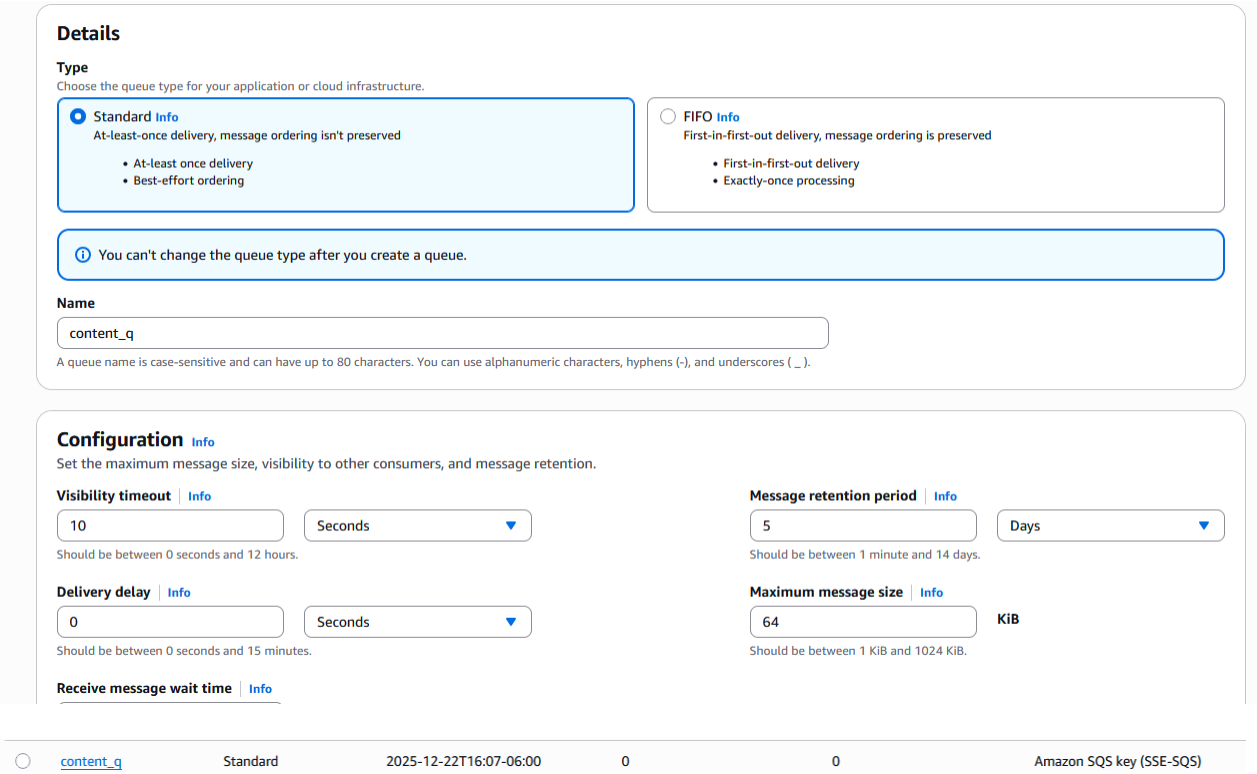


Step 2: SQS Queue Setup

Next, an SQS queue was created through the SQS Management Console. A Standard queue type was chosen, and the queue was named content_q. Several settings were configured:

- Visibility timeout: 10 seconds

- Message retention period: 5 days (note: although the referenced PDF suggested 15 minutes, 5 days was used)
- Maximum message size: 64 KB

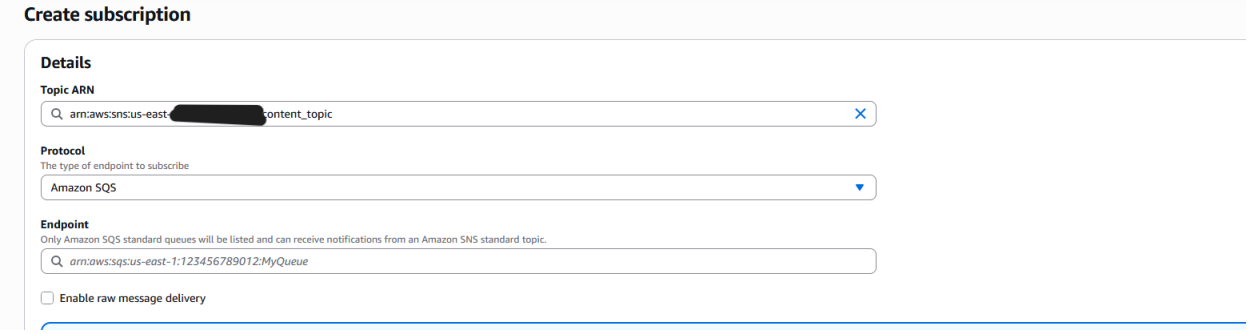


Step 3: Queue Verification

After creation, the queue content_q appeared in the list of available queues. It was confirmed to have been created on 12/22/2025 at 4:07 PM (UTC-6). At this point, there were zero messages available in the queue. The queue was encrypted using the Amazon SQS key (SSE-SQS).

Step 4: Establishing the SNS to SQS Subscription

To connect the SNS topic to the SQS queue, a subscription was created. Returning to the SNS content_topic, it was noted that there were no existing subscriptions. A new subscription was initiated by selecting the Amazon SQS protocol and choosing the ARN of the content_q queue as the endpoint.

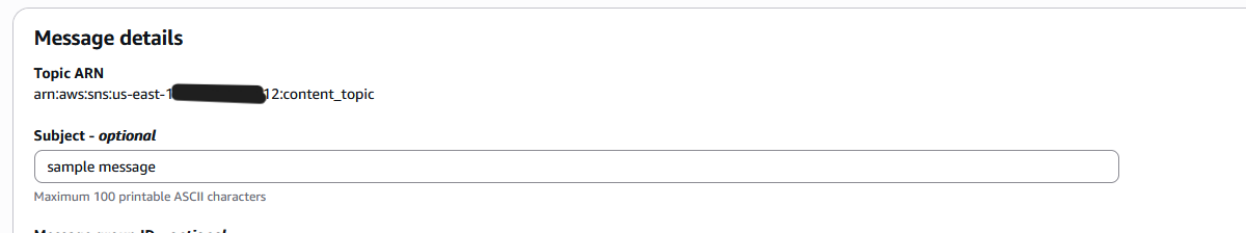


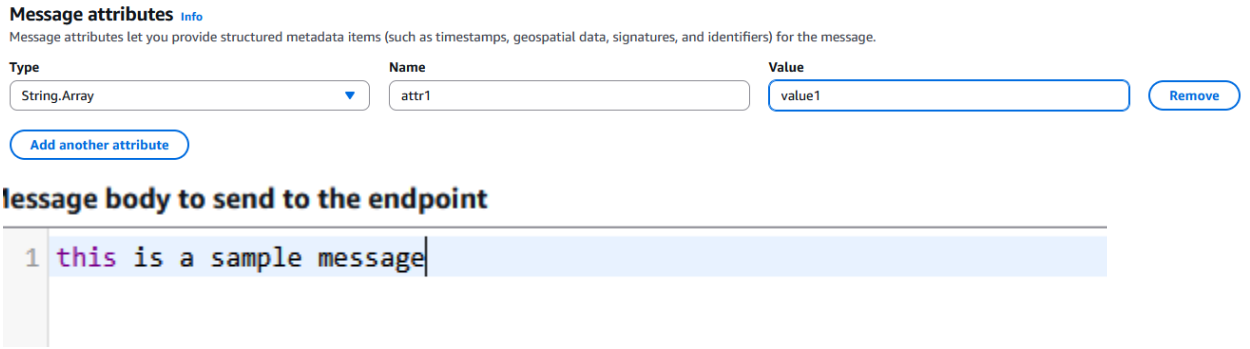
Step 5: Publishing a Test Message

A test message was published to verify the integration. The subject of the message was "sample message," with the body containing "this is a sample message." A message attribute was added with the following details:

- Type: String.Array (the PDF indicated String, but String.Array was used)
- Name: attr1
- Value: value1

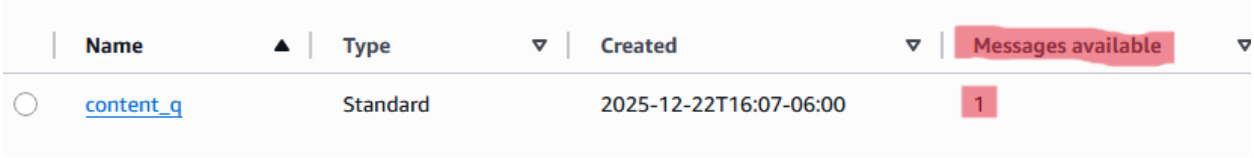
The message was then published to the SNS topic.



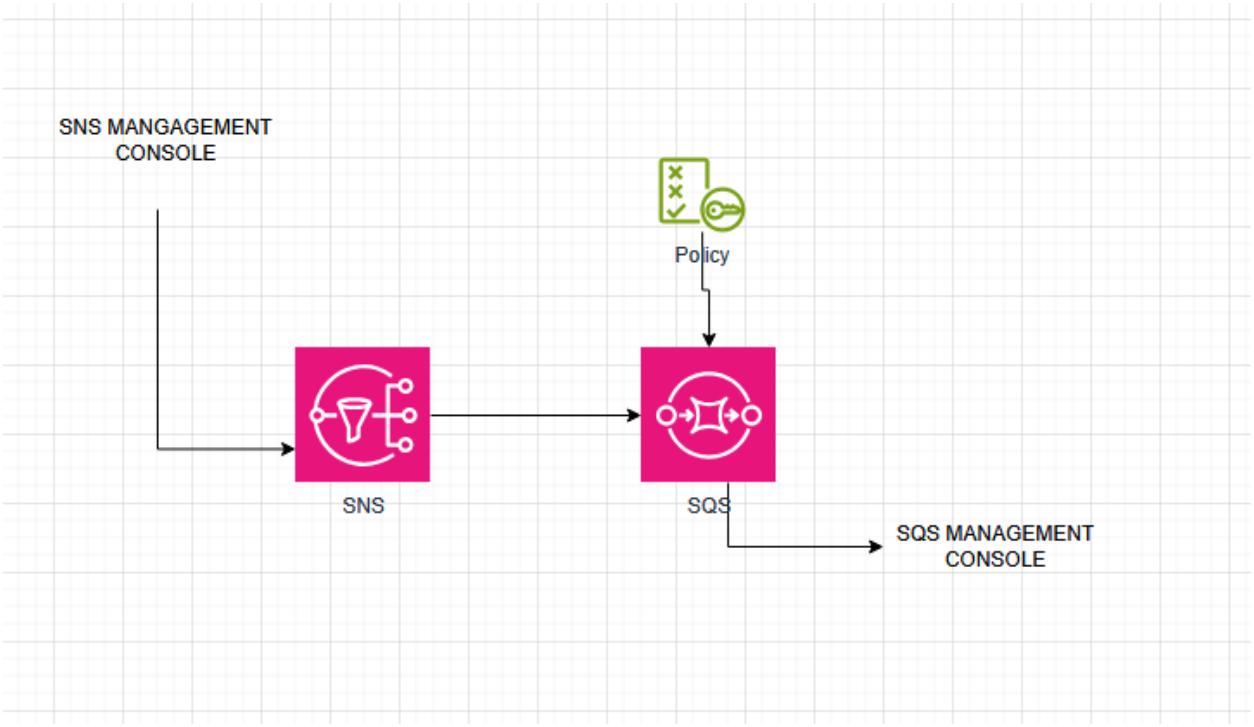


Step 6: Message Delivery Verification

The final step involved confirming that the message was successfully delivered. Upon checking the SQS console, the content_q queue now showed one message available. This verified that the SNS topic had successfully delivered the message to the SQS queue.



Architecture:



Summary

The AWS SNS/SQS integration project was completed successfully. The message flow worked as intended:

SNS Topic (content_topic) → Subscription → SQS Queue (content_q)

This project demonstrated the publish-subscribe messaging pattern, showcasing how SNS can distribute messages to SQS queues for asynchronous processing.

References:

1. Amazon Simple Notification Service (SNS) Documentation

- URL: https://docs.aws.amazon.com/sns/

- Description: Official AWS documentation covering SNS topics, subscriptions, message publishing, and protocols. Essential for understanding how to create and manage SNS topics.

2. Amazon Simple Queue Service (SQS) Documentation

- URL: https://docs.aws.amazon.com/sqs/

- Description: Complete guide to SQS queue types, message handling, visibility timeout, retention periods, and queue configuration. Critical for understanding queue behavior and settings.

3. Fanout to Amazon SQS Queues - SNS Tutorial

- URL: https://docs.aws.amazon.com/sns/latest/dg/sns-sqs-as-subscriber.html

- Description: Step-by-step guide on subscribing SQS queues to SNS topics, including IAM policy requirements and best practices for the fanout pattern used in this project.

4. Amazon SQS Access Policy Examples

- URL: https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-creating-custom-policies-access-policy-examples.html

- Description: Documentation on SQS access policies including examples for granting SNS permission to send messages to queues. Explains the JSON policy structure used in this project.